



GIRRAWEEN HIGH SCHOOL
YEAR 11 YEARLY EXAMINATION

2007

MATHEMATICS

2 UNIT

Time allowed- Two Hours
(plus 5 minutes reading time)

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question.
- Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.
- Start each question on a new page.
- Write your name on each piece of paper.

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Question 1 (18 marks)**Marks**

- (a) Find, correct to 3 significant figures: $\frac{6.4^2 - 3.4^3}{2 \times 6.4 \times 3.4}$ 2
- (b) Factorise fully: 4
- (i) $2x^2 - 7x - 15$ (ii) $12x^2 - 3y^2$
- (c) Find the exact value: 2
- (i) $\frac{3^{-2}}{5^{-3}}$ (ii) $9^{\frac{1}{2}} \times 100^{\frac{-3}{2}}$
- (d) Solve: $3 - (4 - x) = 5x$ 2
- (e) Rationalise the denominator: $\frac{\sqrt{5} - \sqrt{2}}{3\sqrt{2} - \sqrt{5}}$ 4
- (f) Solve and graph on a number line: $|2x - 1| \geq 3$ 4

Question 2 (15 marks)

- (a) The points A(-4, 2), B(1, 3) and C(5, -4) are the vertices of a triangle. 9
- (i) Plot the points on a number plane showing the above information.
- (ii) Find the gradient of AC.
- (iii) Find the equation of AC.
- (iv) Find the equation of the line perpendicular to AC and passing through B.
- (b) Find the perpendicular distance between the parallel lines $3x - y + 1 = 0$ and $3x - y + 5 = 0$. 3
- (c) The points (4, -1), (5, 2) and (7, k) are collinear. Find the value of k . 3

Question 3 (14 marks)

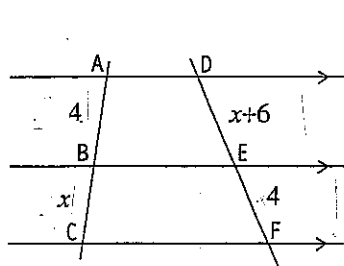
(a) Each interior angle of a regular polygon is 162° . Find

3

- (i) the number of sides.
- (ii) the sum of its interior angles.

(b) Find the value of x . Give reasons.

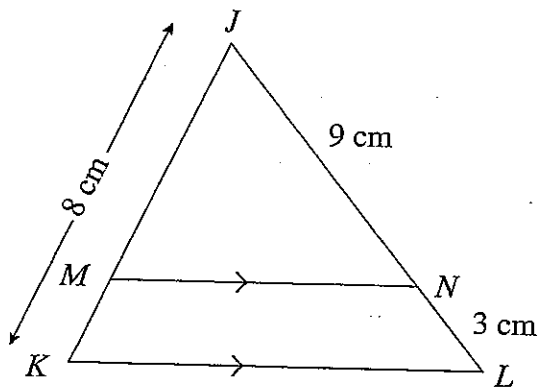
3



(c) In the diagram given below, $\triangle JMN$ is similar to $\triangle JKL$.

Find the length of MK .

2



(d) Sketch the region:

6

(i) $y < \sqrt{16 - x^2}$

(ii) $y \leq 2 - x^2$ and $y \geq |x|$

Question 4 (28 marks)(a) For each of the following curves 16

$(\alpha) y = (x+1)^2$

$(\beta) y = |x| - 3$

$(\gamma) y = \frac{1}{x-4}$

$(\delta) y = \sqrt{2-x}$

(i) Draw a neat sketch showing important features.

(ii) State the domain and range.

(b) Find the domain of $y = \frac{1}{\sqrt{25-x^2}}$. 2(c) Is the function $f(x) = \frac{x}{x^2-1}$ odd, even or neither? Justify your answer. 3

(d) A function is defined by the rule

$$f(x) = \begin{cases} x^2 - 1 & x \geq -1 \\ x + 1 & x < -1 \end{cases}$$

(i) Sketch the curve. 3(ii) State the domain and range. 2(iii) Find the value of $f(2) + f(-2)$. 2**Question 5 (23 marks)**(a) Differentiate: 11

$(i) y = 3\sqrt[3]{x} + 5x - 4$

$(ii) y = (7x^3 - 1)(5x^2 - x)$

$(iii) y = \frac{4x^2 - 2}{x^2 + 5}$

$(iv) y = \sqrt{x^2 - 2x}$

(b) Find the derivative of $y = \frac{1}{x}$ from first principles. 3(c) Find the equation of the normal to the parabola $y = 2x^2 - 5x + 1$ at the point on the parabola where the gradient is 3. 4(d) Evaluate the following limits: 5

$(i) \lim_{x \rightarrow 5} \frac{x^3 - 125}{x^2 - 7x + 10}$

$(ii) \lim_{x \rightarrow \infty} \frac{x^2 + 1}{2x^2 + 5}$

Question 6 (19 marks)

(a) The quadratic equation $4x^2 - 9x + 1 = 0$ has roots α and β . Find: 10

(i) $\alpha + \beta$ (ii) $\alpha\beta$ (iii) $\alpha^2 + \beta^2$ (iv) $(\alpha + 2)(\beta + 2)$

(b) Find the values of k for which $y = kx^2 + 3kx + 6$ is positive definite. 3

(c) The line $y = 3x - p + 1$ is a tangent to the parabola $y = x^2$. Find the value of p . 3

(d) Find the values of A, B, C if

$$4x^2 - 9x - 2 \equiv Ax(x+1) + B(x-1) + C \text{ for all } x. \quad 3$$

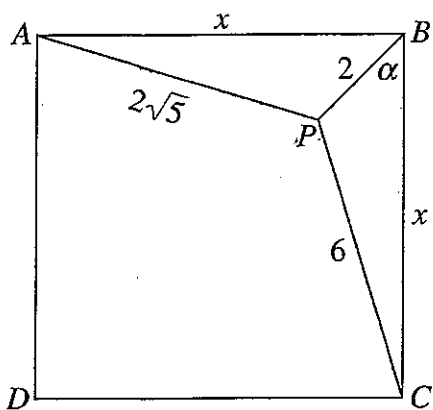
Question 7 (21 marks)

(a) Solve for $0^\circ \leq \theta \leq 360^\circ$. Write answer correct to the nearest minute. 6

(i) $2\sin\theta = \sqrt{3}$ (ii) $\cos 2\theta = \frac{-1}{\sqrt{2}}$

(b) Prove: $(\cot\theta + \operatorname{cosec}\theta)^2 = \frac{1 + \cos\theta}{1 - \cos\theta}$ 3

(c) The diagram shows a square $ABCD$ of side x cm. $PC = 6$ cm, $PB = 2$ cm and $AP = 2\sqrt{5}$ cm.



(i) Using the cosine rule in $\triangle PBC$, show that $\cos\alpha = \frac{x^2 - 32}{4x}$. 2

(ii) By considering $\triangle PBA$, show that $\sin\alpha = \frac{x^2 - 16}{4x}$. 2

(iii) Show that the value of x is a solution of $x^4 - 56x^2 + 640 = 0$ (Hint: $\sin^2\alpha + \cos^2\alpha = 1$). 3

(iv) Solve $x^4 - 56x^2 + 640 = 0$ and determine the length AB of the square. Give reason. 5

Years 11 Mathematics Yearly 2007 - Solutions

Question 1

(a) 0.0381 (2)

(b) (i) $2x^2 - 7x - 15$

$= (x-5)(2x+3)$ (2)

(ii) $12x^2 - 3y^2$

$= 3(4x^2 - y^2)$

$= 3(2x+y)(2x-y)$ (2)

(c) (i) $\frac{3^{-2}}{5^{-3}} = \frac{1}{9} \times \frac{125}{1}$

$= \frac{125}{9}$

$= 13\frac{8}{9}$ (1)

(ii) $9^{\frac{1}{2}} \times 100^{-\frac{3}{2}}$

$= 3 \times \frac{1}{100^{\frac{3}{2}}}$ (1)

$= \frac{3}{1000}$

(d) $3 - (4 - x) = 5x$

$3 - 4 + x = 5x$

$4x = -1$ (2)

$x = -\frac{1}{4}$

(e) $\frac{\sqrt{5} - \sqrt{2}}{3\sqrt{2} - \sqrt{5}} = \frac{(\sqrt{5} - \sqrt{2})(3\sqrt{2} + \sqrt{5})}{(3\sqrt{2} - \sqrt{5})(3\sqrt{2} + \sqrt{5})}$

$= \frac{3\sqrt{10} + 5 - 6 - \sqrt{10}}{(3\sqrt{2})^2 - (\sqrt{5})^2}$

$= \frac{2\sqrt{10} - 1}{18 - 5} = \frac{2\sqrt{10} - 1}{13}$ (4)

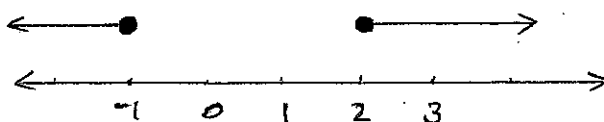
(f) $|2x-1| \geq 3$

$2x-1 \geq 3$ or $-(2x-1) \geq 3$

$2x-1 \geq 3$ or $2x-1 \leq -3$

$2x \geq 4$ or $2x \leq -2$ (4)

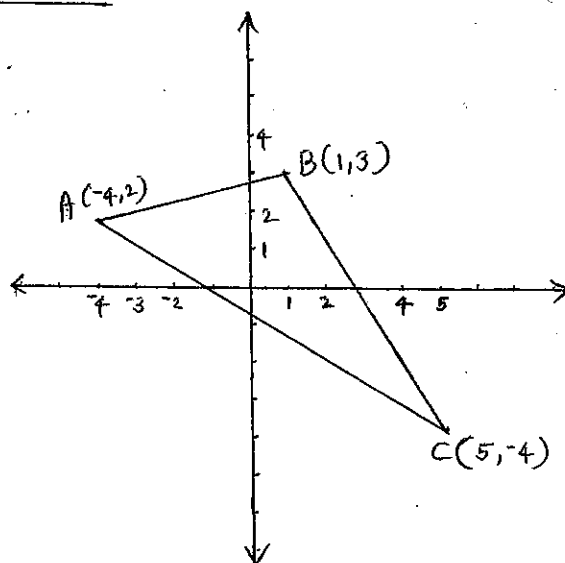
$x \geq 2$ or $x \leq -1$



Question 2

(a)

(i)



(ii) $m_{AC} = \frac{2+4}{-4-5} = \frac{6}{-9} = -\frac{2}{3}$ (2)

(iii) $y-2 = -\frac{2}{3}(x+4)$

$3y-6 = -2(x+4)$ (3)

$3y-6 = -2x-8$

$2x+3y+2=0$ OR $y = -\frac{2}{3}x - \frac{2}{3}$

(IV) Gradient of the perpendicular = $\frac{3}{2}$

Equation of the perpendicular line is

$$y - 3 = \frac{3}{2}(x - 1)$$

$$2y - 6 = 3(x - 1)$$

$$2y - 6 = 3x - 3$$

(3)

$$3x - 2y + 3 = 0$$

OR

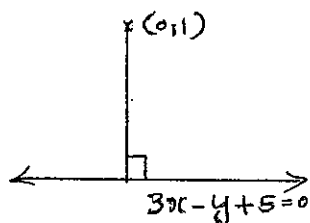
$$y = \frac{3}{2}x + \frac{3}{2}$$

(b) $3x - y + 1 = 0$

when $x = 0$, $1 - y = 0$

$1 = y$

$(0, 1)$ is a point on $3x - y + 1 = 0$



(3)

$$d = \frac{|3 \times 0 - 1 + 5|}{\sqrt{3^2 + (-1)^2}}$$

$$= \frac{4}{\sqrt{10}} \text{ OR } \frac{2\sqrt{10}}{5}$$

(c) $A(4, -1)$ $B(5, 2)$ $C(7, k)$

$$m_{AB} = m_{BC}$$

$$3 = \frac{k - 2}{2}$$

(3)

$$k - 2 = 6$$

$$k = 8$$

Question 3

(a) (i) Let n be the number of sides

$$\frac{(n-2) \times 180}{n} = 162$$

$$180(n-2) = 162n$$

$$180n - 360 = 162n$$

$$18n = 360$$

$$n = 20$$

(2)

(ii) Sum of interior angles = $(n-2) \times 180$

$$= 18 \times 180$$

$$= 3240$$

(1)

(b) $\frac{4}{x} = \frac{x+6}{4}$ (A family of parallel lines cutting transversals, preserves the ratio of sides on each transversal)

$$x(x+6) = 16$$

$$x^2 + 6x - 16 = 0$$

$$(x+8)(x-2) = 0$$

$$x = -8 \text{ OR } 2$$

$x = 2$ (negative values can't be the length)

(3)

(c) $\frac{JM}{JK} = \frac{JN}{JL}$

$$\frac{JM}{8} = \frac{9}{12}$$

$$JM = \frac{9 \times 8}{12} = 6$$

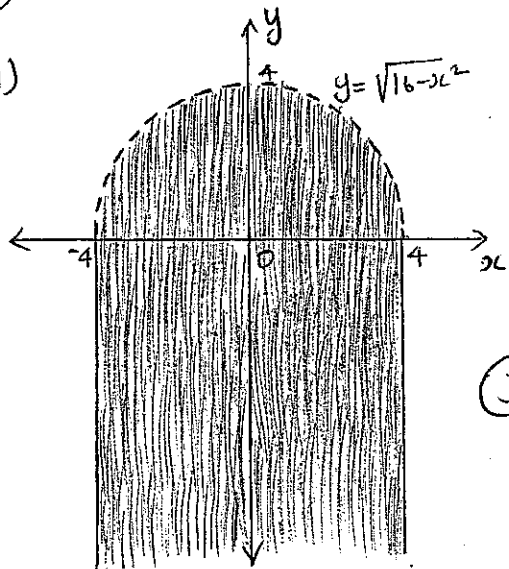
(2)

$$MK = 8 - 6$$

$$= 2 \text{ cm}$$

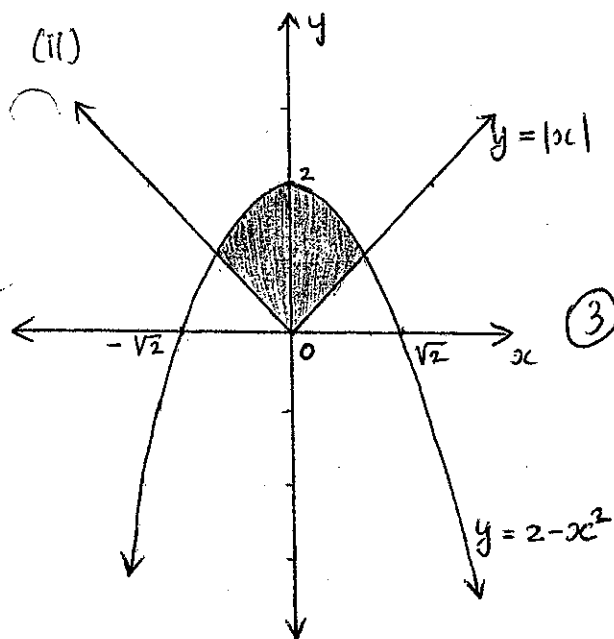
(d)

(i)



③

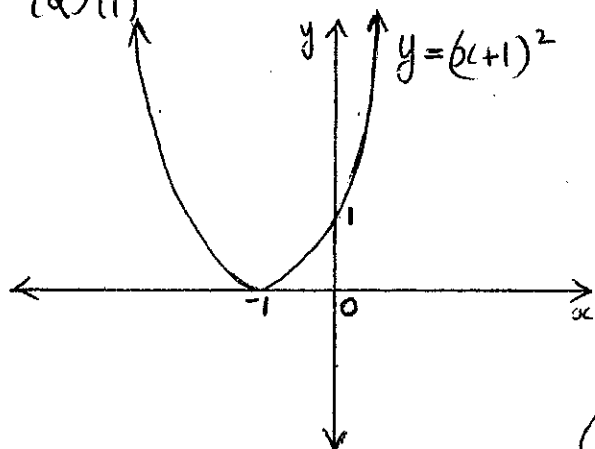
(ii)



③

Question 4

(a)(i)



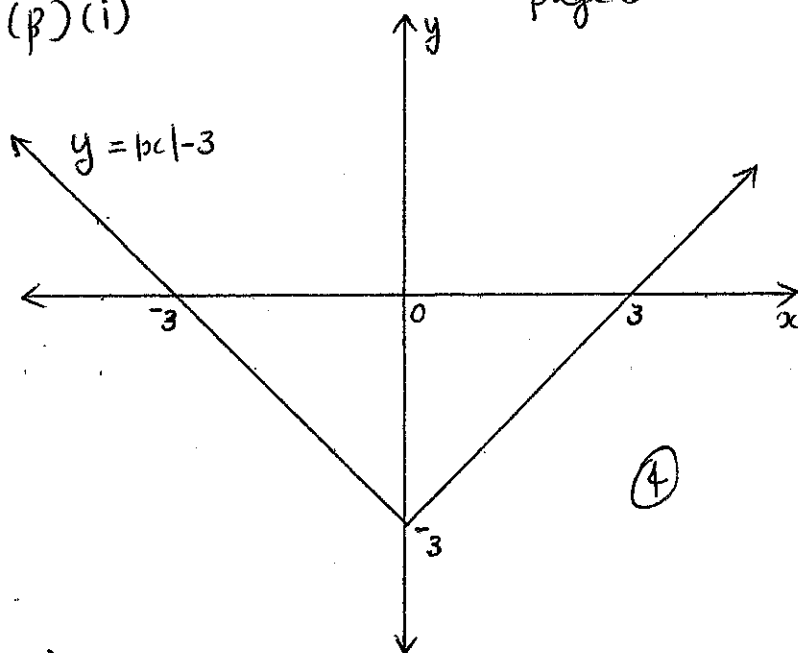
④

(ii) D : all real x

R : $y \geq 0$, y real

(p)(i)

page 3

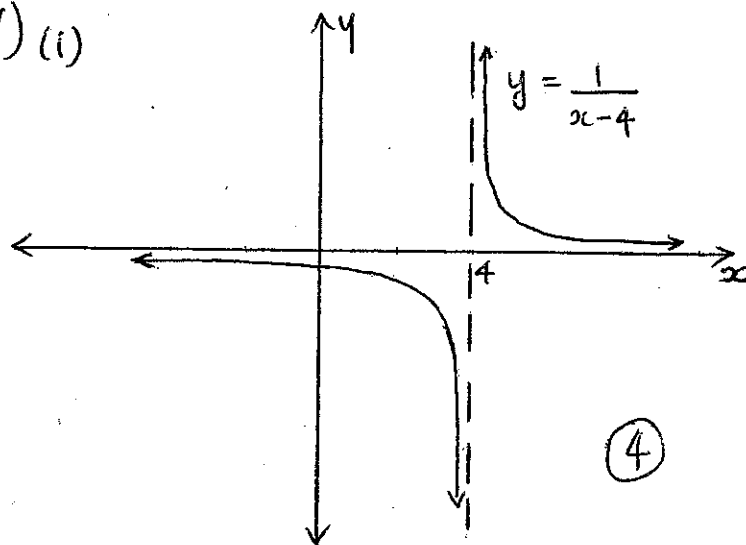


④

(ii) D : all real x

R : $y \geq 3$, $y \in \mathbb{R}$

(q)(i)

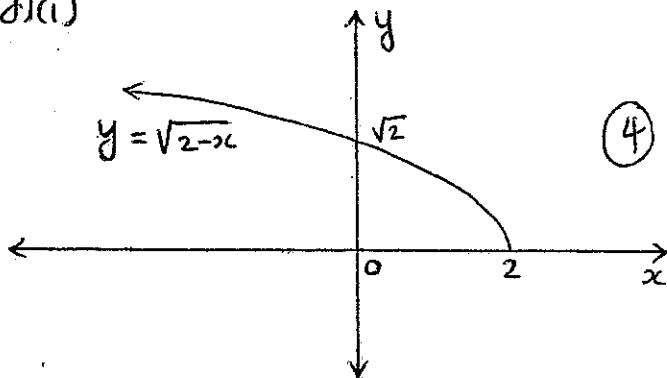


④

(ii) D : all real x , $x \neq 4$

R : all real y , $y \neq 0$

(r)(i)



④

(ii) D : $x \leq 2$

R : $y \geq 0$

$$(b) y = \frac{1}{\sqrt{25-x^2}}$$

$\sqrt{25-x^2}$ exists when

$$25-x^2 \geq 0$$

$$25 \geq x^2$$

$$x^2 \leq 25$$

$$\text{ie } -5 \leq x \leq 5$$

$$D : \underline{-5 < x < 5}$$

$$(c) f(-x) = \frac{-x}{(x)^2-1}$$

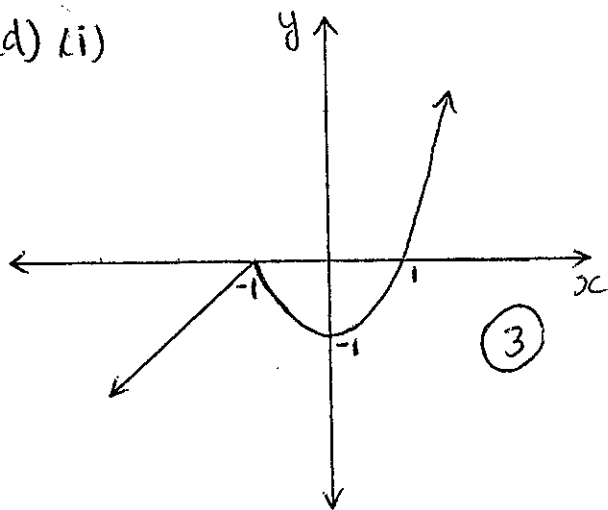
$$= \frac{-x}{x^2-1}$$

$$= -\left(\frac{x}{x^2-1}\right)$$

$$= -f(x)$$

$\therefore \underline{f(x) \text{ is an odd function.}}$

(d) (i)



(ii) Domain : $\mathbb{R}$

Range : $\mathbb{R}$

$$(iii) f(2) + f(-2)$$

$$= (4-1) + (-2+1)$$

$$= \underline{\underline{3-1=2}}$$

page 4

Question 5

$$(a) (i) y = 3(x)^{\frac{1}{3}} + 5x - 4$$

$$y' = 3 \times \frac{1}{3} x^{\frac{1}{3}-1} + 5 \quad (2)$$

$$= x^{-\frac{2}{3}} + 5 = \underline{\underline{\sqrt[3]{x^{-2}} + 5}}$$

$$(ii) y = (7x^3-1)(5x^2-x)$$

$$y' = (7x^3-1)(10x-1) + (5x^2-x)21x^2$$

$$= \underline{\underline{175x^4 - 28x^3 - 10x + 1}} \quad (3)$$

$$(iii) y = \frac{4x^2-2}{x^2+5}$$

$$y' = \frac{(x^2+5)8x - (4x^2-2)2x}{(x^2+5)^2}$$

$$= \frac{8x^3 + 40x - 8x^3 + 4x}{(x^2+5)^2}$$

$$= \frac{44x}{(x^2+5)^2}$$

$$= \underline{\underline{\frac{44x}{(x^2+5)^2}}}$$

$$(iv) y = \sqrt{x^2-2x}$$

$$y' = \frac{1}{2\sqrt{x^2-2x}} \times 2x-2 = \frac{2(x-1)}{2\sqrt{x^2-2x}}$$

$$= \underline{\underline{\frac{x-1}{\sqrt{x^2-2x}}}}$$

$$(b) f(x) = \frac{1}{x} \quad f(x+h) = \frac{1}{x+h}$$

$$f(x+h) - f(x) = \frac{1}{x+h} - \frac{1}{x}$$

$$= \frac{x - x-h}{x(x+h)} = \frac{-h}{x(x+h)}$$

$$\frac{f(x+h) - f(x)}{h} = \frac{-1}{x(x+h)}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{-1}{x(x+h)} \quad (3)$$

$$= \frac{-1}{x^2}$$

$$(c) y = 2x^2 - 5x + 1$$

$$y' = 4x - 5$$

$$4x - 5 = 3$$

$$x = 2$$

$$\text{When } x = 2, y = -1$$

Equation of normal

$$y + 1 = \frac{-1}{3}(x - 2)$$

$$3y + 3 = -x + 2$$

$$\underline{x + 3y + 1 = 0}$$

$$(d) (i) \lim_{x \rightarrow 5} \frac{x^3 - 125}{x^2 - 7x + 10}$$

$$= \lim_{x \rightarrow 5} \frac{(x-5)(x^2 + 5x + 25)}{(x-5)(x-2)}$$

$$= \lim_{x \rightarrow 5} \frac{x^2 + 5x + 25}{x-2} \quad (3)$$

$$= \underline{25}$$

$$(ii) \lim_{x \rightarrow \infty} \frac{x^2 + 1}{2x^2 + 5}$$

$$= \lim_{x \rightarrow \infty} \frac{1 + \frac{1}{x^2}}{2 + \frac{5}{x^2}} \quad (2)$$

$$= \underline{\frac{1}{2}} \quad \left(\because \lim_{x \rightarrow \infty} \frac{1}{x} = 0 \right)$$

Question 6

page 5

$$(a) 4x^2 - 9x + 1 = 0$$

$$(i) \alpha + \beta = \frac{-b}{a} \quad (2)$$

$$= \frac{-(-9)}{4} = \frac{9}{4}$$

$$(ii) \alpha\beta = \frac{c}{a} \quad (2)$$

$$= \frac{1}{4}$$

$$(iii) \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= \left(\frac{9}{4}\right)^2 - 2 \times \frac{1}{4}$$

$$= \frac{73}{16} \quad (3)$$

$$(iv) (\alpha+2)(\beta+2) = \alpha\beta + 2(\alpha+\beta) + 4$$

$$= \frac{1}{4} + 2 \times \frac{9}{4} + 4$$

$$= \frac{35}{4} = 8\frac{3}{4} \quad (3)$$

$$(b) y = kx^2 + 3kx + 6 \text{ is positive definite}$$

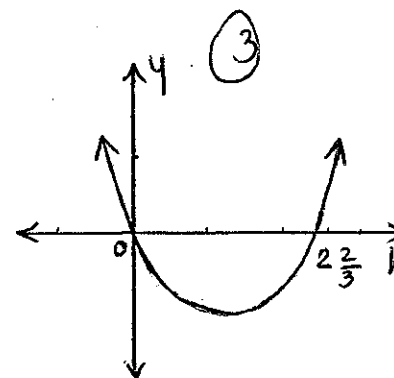
$$\Rightarrow a > 0 \text{ and } \Delta < 0$$

$$\therefore k > 0 \text{ and } \Delta = (3k)^2 - 4k \times 6 < 0$$

$$9k^2 - 24k < 0$$

$$k(9k - 24) < 0$$

$$\text{ie } \underline{0 < k < 2\frac{2}{3}}$$



$$(c) y = x^2 \quad \text{--- (1)}$$

$$y = 3x - p + 1 \quad \text{--- (2)}$$

Solving (1) and (2) simultaneously we

$$\text{get } x^2 - 3x + p - 1 = 0$$

Given that (2) is a tangent to (1)

$$\therefore \Delta = 0$$

$$\Delta = 9 - 4(p-1)$$

$$= 13 - 4p$$

$$13 - 4p = 0$$

$$p = \underline{3\frac{1}{4}} \quad (3)$$

$$(d) 4x^2 - 9x - 2 \equiv Ax(x+1) + B(x-1) + C$$

$$4x^2 - 9x - 2 \equiv Ax^2 + Ax + Bx - B + C$$

Equating coefficients of x^2 , x and constant term we get

$$A = 4$$

$$A + B = -9$$

$$B = -13$$

$$-B + C = -2$$

$$C = -2 + B$$

$$= -2 + -13$$

$$= -15$$

$$\underline{A = 4, B = -13, C = -15}$$

Question 7

$$(a) (i) 2\sin\theta = \sqrt{3}$$

$$\sin\theta = \frac{\sqrt{3}}{2} \quad (2)$$

$$\underline{\theta = 60^\circ, 120^\circ}$$

$$(ii) \cos 2\theta = -\frac{1}{\sqrt{2}}$$

$$\text{Let } u = 2\theta$$

$$\cos u = -\frac{1}{\sqrt{2}}, \quad 0 \leq u \leq 720^\circ$$

$$u = 135^\circ, 225^\circ, 495^\circ, 585^\circ \quad (4)$$

$$\underline{\theta = 67^\circ 30', 112^\circ 30', 247^\circ 30', 292^\circ 30'}$$

$$(b) \text{LHS} = (\cot\theta + \operatorname{cosec}\theta)^2$$

$$= \left(\frac{\cos\theta}{\sin\theta} + \frac{1}{\sin\theta} \right)^2 \quad (3)$$

$$= \frac{(1 + \cos\theta)^2}{\sin^2\theta} = \frac{(1 + \cos\theta)^2}{1 - \cos^2\theta}$$

$$= \frac{(1 + \cos\theta)(1 + \cos\theta)}{(1 + \cos\theta)(1 - \cos\theta)} = \frac{1 + \cos\theta}{1 - \cos\theta}$$

$$= \text{RHS}$$

$$(c) (i) \text{ In } \triangle PBC \quad \text{page 6}$$

$$\cos\alpha = \frac{x^2 + 2^2 - 6^2}{2 \times x \times 2}$$

$$= \frac{x^2 + 4 - 36}{4x} \quad (2)$$

$$= \frac{x^2 - 32}{4x}$$

$$(ii) \text{ In } \triangle PBA$$

$$\cos(90^\circ - \alpha) = \frac{x^2 + 2^2 - (2\sqrt{5})^2}{2 \times x \times 2}$$

$$= \frac{x^2 + 2^2 - 20}{4x}$$

$$= \frac{x^2 - 16}{4x} \quad (2)$$

$$\cos(90^\circ - \alpha) = \sin\alpha$$

$$\sin\alpha = \frac{x^2 - 16}{4x}$$

$$(iii) \sin^2\alpha + \cos^2\alpha = 1$$

$$\left(\frac{x^2 - 32}{4x} \right)^2 + \left(\frac{x^2 - 16}{4x} \right)^2 = 1$$

$$\frac{x^4 - 64x^2 + 1024 + x^4 - 32x^2 + 256}{16x^2} = 1$$

$$2x^4 - 96x^2 + 1280 = 16x^2$$

$$2x^4 - 112x^2 + 1280 = 0 \quad (3)$$

$$\underline{x^4 - 56x^2 + 640 = 0}$$

$$(IV) x^4 - 56x^2 + 640 = 0$$

$$\text{Let } u = x^2$$

$$u^2 - 56u + 640 = 0$$

$$u = \frac{56 \pm \sqrt{3136 - 4 \times 1 \times 640}}{2}$$

$$= \frac{56 \pm \sqrt{576}}{2}$$

$$= \frac{56 \pm 24}{2}$$

$$= 40 \text{ or } 16$$

(5)

$$x^2 = 40 \text{ or } x^2 = 16$$

$$x = \pm\sqrt{40} \text{ or } x = \pm 4$$

$$x = \sqrt{40} \text{ or } x = 4 \text{ (} x \text{ can't be negative)}$$

When $x = 4$, $\triangle PBC$ does not exist (The sum of two sides of a $\triangle$ must be greater than the third side)

$$\therefore x = \underline{\underline{\sqrt{40} \text{ cm}}}$$

or

$$= \underline{\underline{2\sqrt{10} \text{ cm}}}$$

